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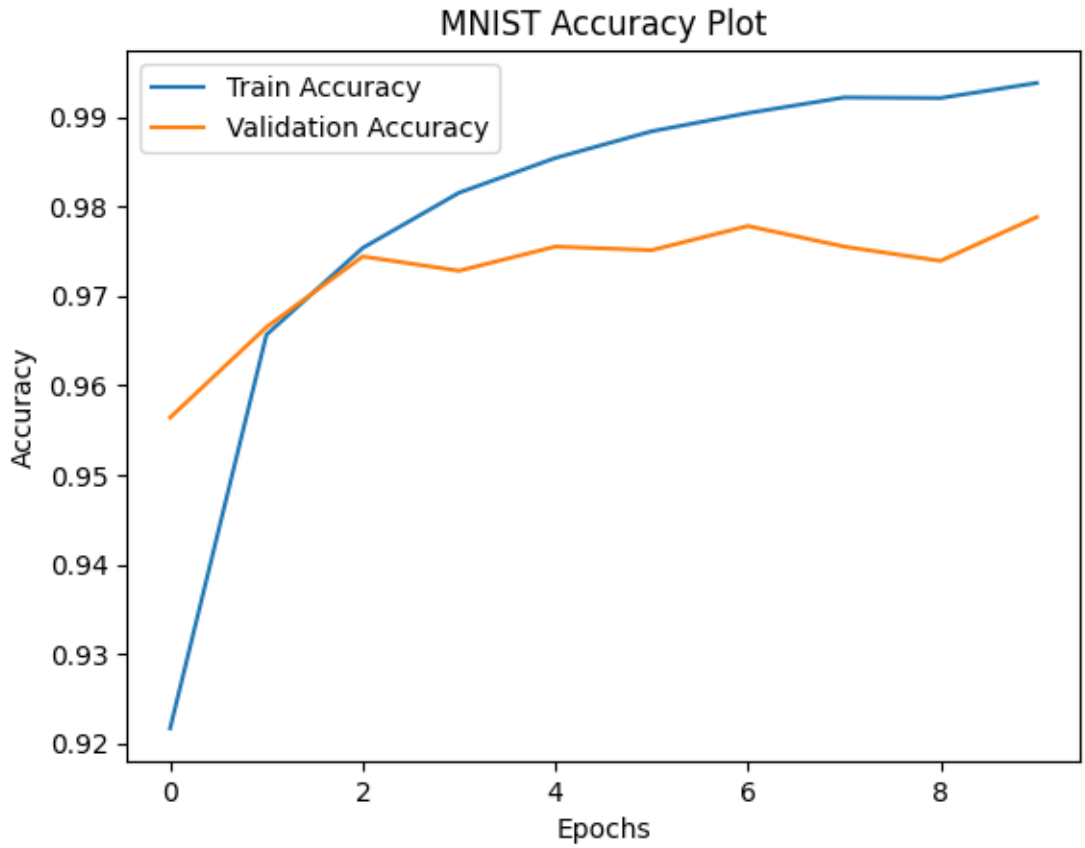
1)MNIST DIGIT CLASSIFICATION

```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, classification_report
import seaborn as sns

# Load dataset
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()

# Normalize images
x_train = x_train / 255.0
x_test = x_test / 255.0

# Reshape for neural network
x_train = x_train.reshape(-1, 28*28)
x_test = x_test.reshape(-1, 28*28)

# Build model
model = tf.keras.models.Sequential([
  tf.keras.layers.Dense(128, activation='relu', input_shape=(784,)),
  tf.keras.layers.Dense(64, activation='relu'),
  tf.keras.layers.Dense(10, activation='softmax')
])

# Compile model
model.compile(optimizer='adam',
```

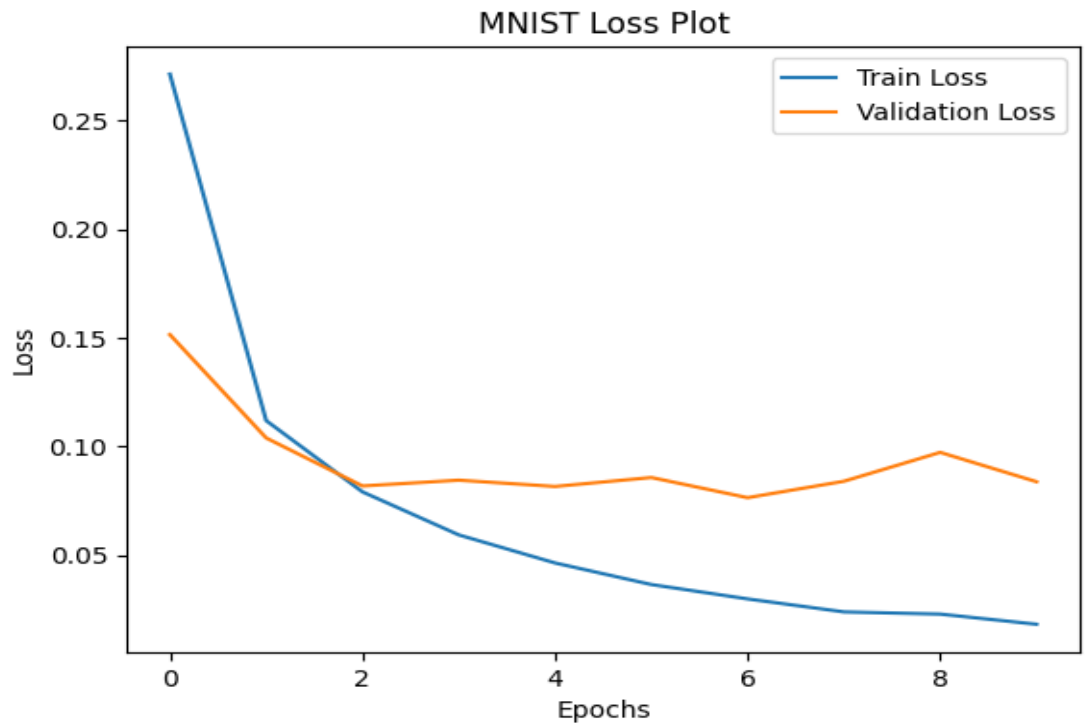
```
        loss='sparse_categorical_crossentropy',
        metrics=['accuracy'])

# Train model
history = model.fit(x_train, y_train,
            validation_data=(x_test, y_test),
            epochs=10, batch_size=64)

# Evaluate model
test_loss, test_acc = model.evaluate(x_test, y_test)
print("\nTest Accuracy:", test_acc)

# Accuracy Plot
plt.figure()
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title("MNIST Accuracy Plot")
plt.xlabel("Epochs")
plt.ylabel("Accuracy")
plt.legend()
plt.show()

#  Loss Plot
plt.figure()
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title("MNIST Loss Plot")
plt.xlabel("Epochs")
```

```
plt.ylabel("Loss")
plt.legend()
plt.show()

# Confusion Matrix
y_pred = np.argmax(model.predict(x_test), axis=1)
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(8,6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title("MNIST Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# Classification Report
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))
```

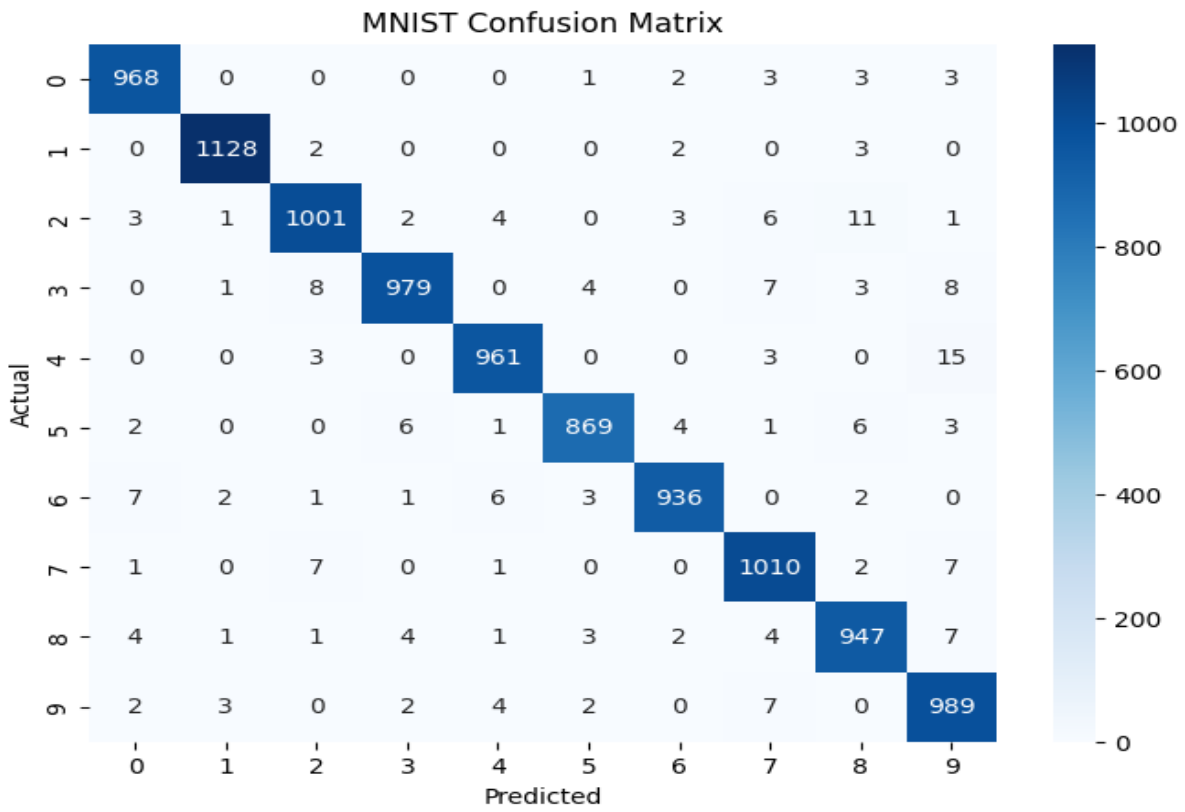
MNIST Accuracy Plot



MNIST Lost Plot



MNIST Confusion Matrix



Classification Report

```
Classification Report:

              precision    recall  f1-score   support

           0       0.98      0.99      0.98       980
           1       0.99      0.99      0.99      1135
           2       0.98      0.97      0.97      1032
           3       0.98      0.97      0.98      1010
           4       0.98      0.98      0.98       982
           5       0.99      0.97      0.98       892
           6       0.99      0.98      0.98       958
           7       0.97      0.98      0.98      1028
           8       0.97      0.97      0.97       974
           9       0.96      0.98      0.97      1009

    accuracy                           0.98     10000
   macro avg       0.98      0.98      0.98     10000
weighted avg       0.98      0.98      0.98     10000
```

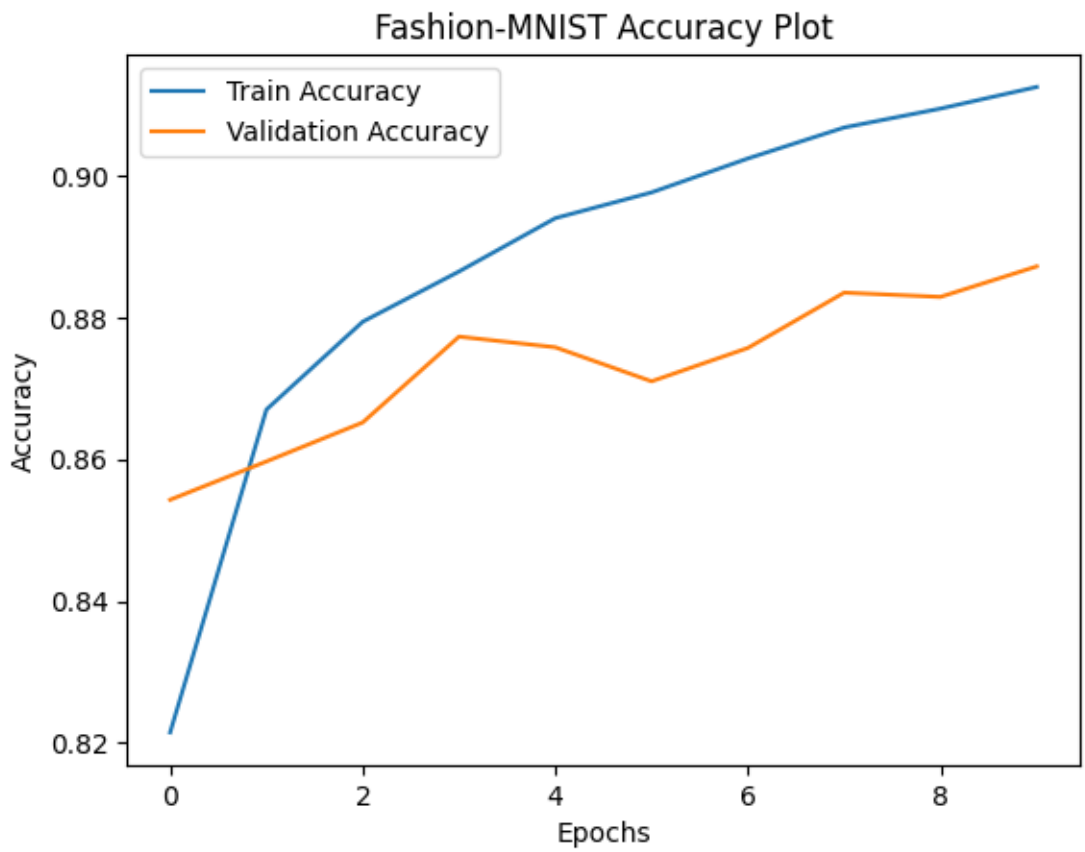
2) FASHION MNIST CLASSIFICATION

```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, classification_report
import seaborn as sns

# Load dataset
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.fashion_mnist.load_data()

# Normalize images
x_train = x_train / 255.0
x_test = x_test / 255.0

# Reshape
x_train = x_train.reshape(-1, 28*28)
x_test = x_test.reshape(-1, 28*28)

# Build model
model = tf.keras.models.Sequential([
  tf.keras.layers.Dense(256, activation='relu', input_shape=(784,)),
  tf.keras.layers.Dense(128, activation='relu'),
  tf.keras.layers.Dense(64, activation='relu'),
  tf.keras.layers.Dense(10, activation='softmax')
])

# Compile model
```

```
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# Train model
history = model.fit(x_train, y_train,
                    validation_data=(x_test, y_test),
                    epochs=10, batch_size=64)

# Evaluate model
test_loss, test_acc = model.evaluate(x_test, y_test)
print("\nTest Accuracy:", test_acc)

# Accuracy Plot
plt.figure()
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title("Fashion-MNIST Accuracy Plot")
plt.xlabel("Epochs")
plt.ylabel("Accuracy")
plt.legend()
plt.show()

# Loss Plot
plt.figure()
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title("Fashion-MNIST Loss Plot")
```

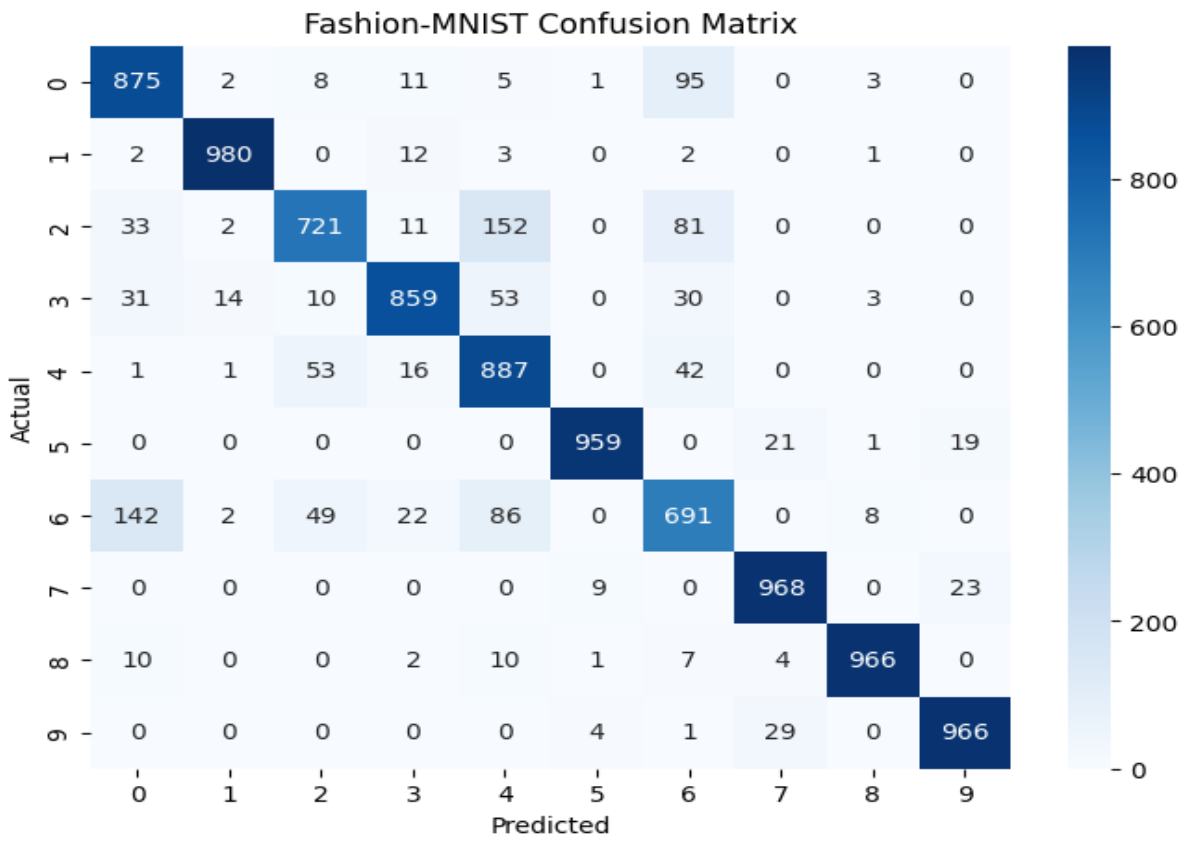
```
plt.xlabel("Epochs")
plt.ylabel("Loss")
plt.legend()
plt.show()

# Confusion Matrix
y_pred = np.argmax(model.predict(x_test), axis=1)
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(8,6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title("Fashion-MNIST Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# Classification Report
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))
```

Fashion MNIST Accuracy Plot



Fashion MNIST Loss Plot

Fashion MNIST Confusion Matrix



Classification Report

```
Classification Report:

              precision    recall  f1-score   support

           0       0.80      0.88      0.84      1000
           1       0.98      0.98      0.98      1000
           2       0.86      0.72      0.78      1000
           3       0.92      0.86      0.89      1000
           4       0.74      0.89      0.81      1000
           5       0.98      0.96      0.97      1000
           6       0.73      0.69      0.71      1000
           7       0.95      0.97      0.96      1000
           8       0.98      0.97      0.97      1000
           9       0.96      0.97      0.96      1000

    accuracy                           0.89     10000
   macro avg       0.89      0.89      0.89     10000
weighted avg       0.89      0.89      0.89     10000
```

Google Colab Link

https://colab.research.google.com/drive/1NHj3A9FvjeohaNnjzlGb_asmJKWIeD1L?usp=sharing